Day 6 Student Handout

Geometry Summer Credit | Triangle Relationships and Coordinate Triangles

Name:	Date:	Period:

Learning Targets

I can use triangle inequalities and side-angle relationships to decide what is possible. I can use midsegments, medians, midpoint, and distance in coordinate triangle problems.

Lesson 6A: Triangle Inequalities and Side-Angle Relationships

Warm-Up

1.	Triangle ABC is congruent to triangle DEF. If AC = 2x + 4 and DF = 18, find x.

2.	Find the missing angle in a triangle with angles 36 degrees and 89 degrees.

3.	Find the distance between A(1, 2) and B(7, 10).

Concept Launch

Can These Sides Make a Triangle?

For any triangle, the sum of the lengths of any two sides must be greater than the third side. If the two shorter sides do not add to more than the longest side, the triangle collapses into a line or cannot be formed.

Problem	Work / notes
Can 4, 7, and 9 form a triangle?	Check shortest + middle > longest: Conclusion:
Can 3, 5, and 9 form a triangle?	Check: Conclusion:

A triangle has side lengths 6, 10, and x. What range of values can x have?	Lower bound: Upper bound: Range:
In a triangle, the angles are 42 degrees, 63 degrees, and 75 degrees. Which side is longest?	Largest angle: Longest opposite side:

Key Relationship

Sides and Angles Match by Size

The longest side is opposite the largest angle. The shortest side is opposite the smallest angle. This is a ranking relationship, not a congruence shortcut.

4.	Order the sides from shortest to longest if angle A = 35 degrees, angle B = 80 degrees, and angle C = 65 degrees.

5.	Order the angles from smallest to largest if AB = 9, BC = 14, and AC = 11.

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Lesson 6B | Midsegments, Medians, and Coordinate Triangles

Lesson 6B: Coordinate Triangle Tools

Three Tools

Midpoint finds the middle of a segment. Distance finds side length on the coordinate plane. A midsegment connects midpoints of two sides of a triangle and is parallel to the third side with half its length.

Problem	Work / notes
Find the midpoint of A(-2, 5) and B(6, -1).	Midpoint formula: Answer:
Find the length of segment CD with C(-3, 4) and D(5, 10).	Distance formula: Answer:
A triangle has a side of length 18. A midsegment parallel to that side has what length?	Relationship: Answer:
A triangle has a midsegment of length 7.5. The parallel side has what length?	Relationship: Answer:
A median connects a vertex to the midpoint of the opposite side. If B(2, 8), C(10, 4), and A is the opposite vertex, find the point where the median from A hits BC.	Midpoint of BC: Median endpoint:

Practice

6.	Can 8, 11, and 21 form a triangle? Explain.

7.	Find the range for x if triangle sides are 5, 12, and x.

8.	Find the midpoint of P(4, -6) and Q(-2, 10).

9.	A midsegment is 13 units long. How long is the parallel side?